Predicting Sepsis Onset

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DATA SCIENCE FOR BIOMEDICAL INFORMATICS

INTRODUCTION

PROBLEM

Sepsis is a **life threatening condition** in which the body issues a systemic response to infection. Without timely intervention, the body begins to shut down, leading to total organ failure (severe sepsis), hypotension (septic shock), and death in only **24-48 hours.**

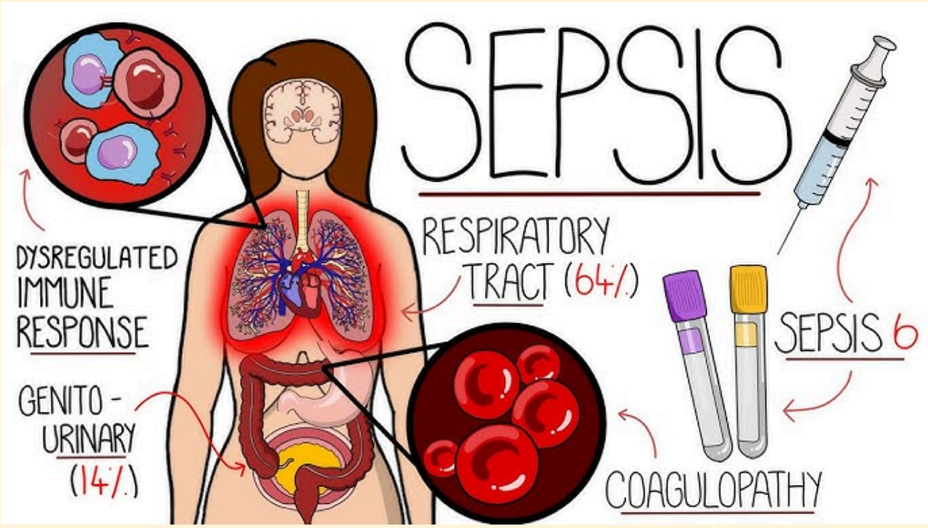
Thus, early prediction of sepsis is vital to saving lives not only because of its mortality rate but also because of its rapid progression.

Treatment, when given immediately, can measurably **increase the chances of survival**, thus saving the patient and reducing the burden of cost and resources on the hospital and its staff.

OBJECTIVE

Our goal is the **early detection of sepsis** using the patient's lab values and vital signs, designing a model that identifies a patient's risk of sepsis and makes a positive or negative (binary) prediction of sepsis for each ICU hour.

We aim to flag sepsis onset at least **3 hours in advance**.



DATA EXPLORATION

DATASET DESCRIPTION

DATASET

- PhysioNet Computing in Cardiology Challenge 2019 dataset. 40,336 ICU patients with 40 clinical variables per hour.
- Two training sets (BIDMC and Emory). The official challenge utility score is used for evaluation.

SOURCE

- Open Database License
- Data obtained from three U.S hospitals and their Electronic Medical Record. Data was deindentified and collected with approval from the IRB.

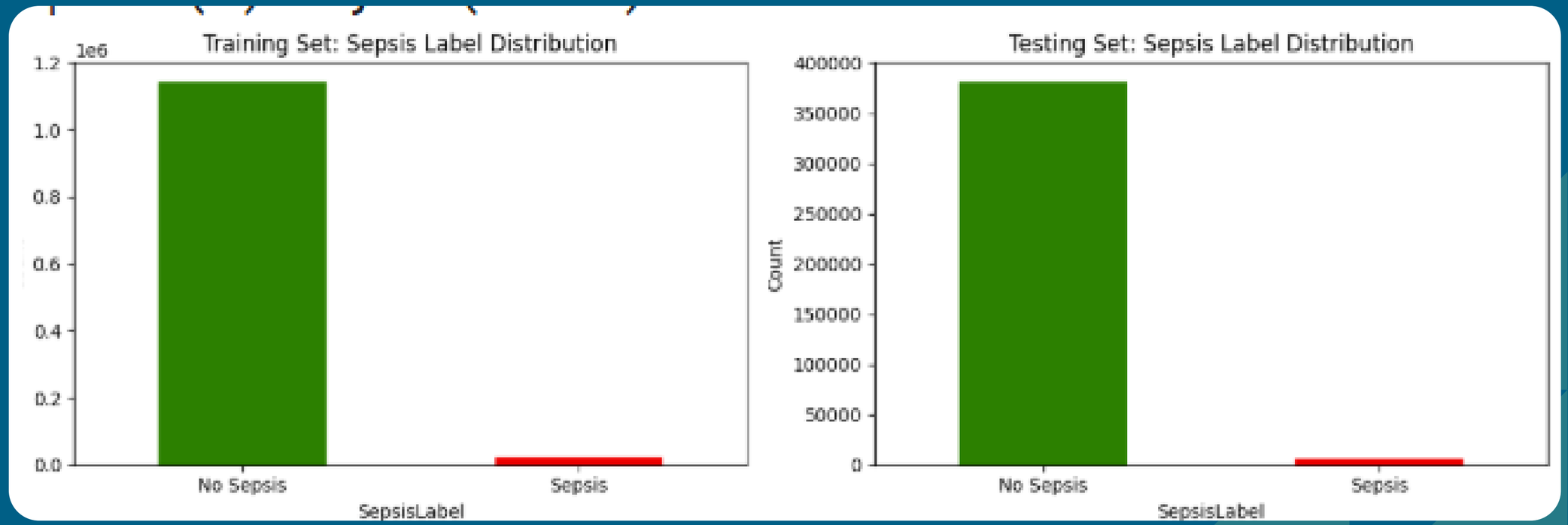
FEATURES

40 CLINICAL FEATURES

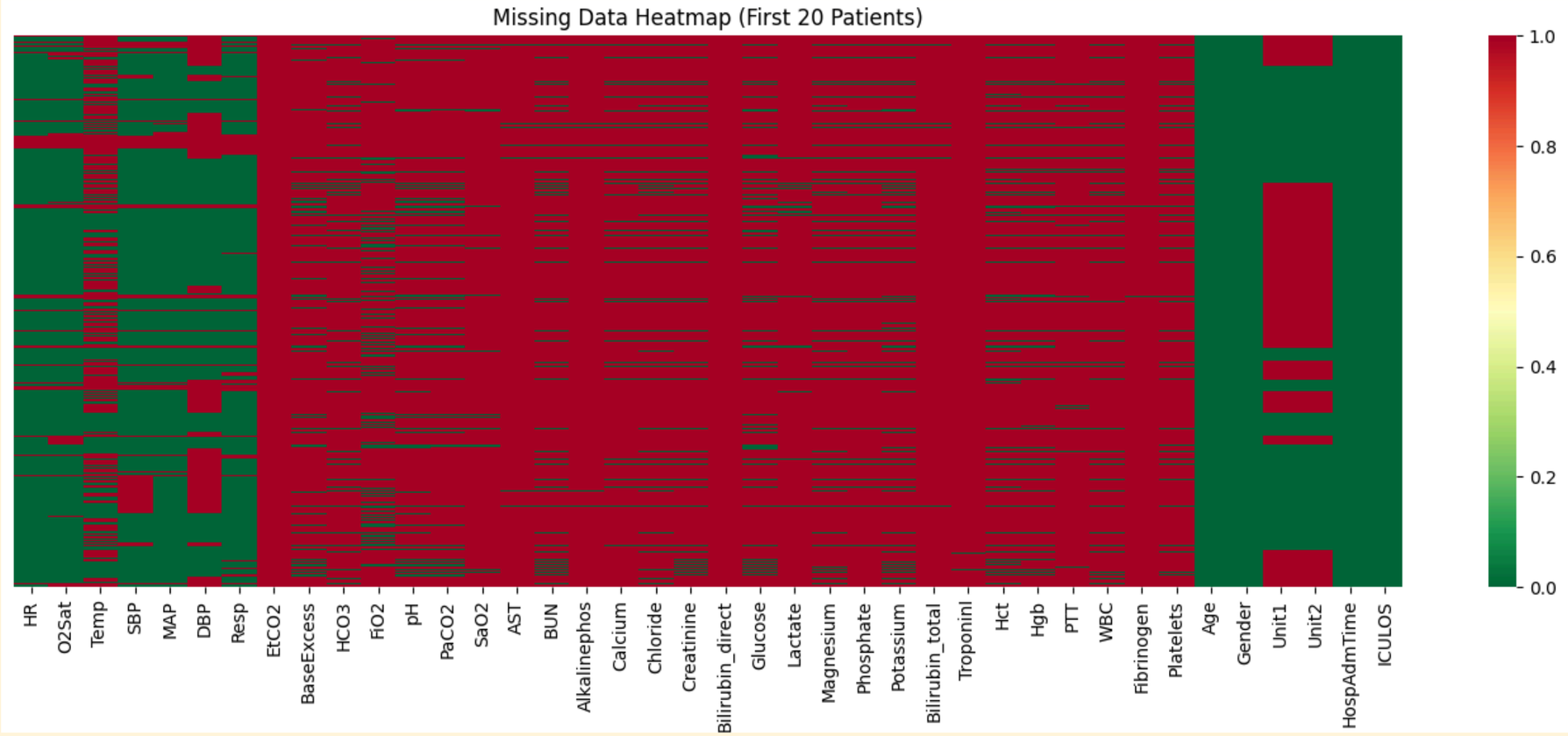
- **8 vital signs** (HR, O2Sat, etc.)
- **26 laboratory variables** (HCO3, pH, Calcium, Hemoglobin, etc.)
- **6 demographic variables** (Age, Gender, etc.)
- **Sepsis label** (1=sepsis, 0= no sepsis)
- 2.5 million hourly time windows and 15 million data points in total

CLASS IMBALANCE

Both Testing and Training: about 98% non-sepsis and 2% sepsis

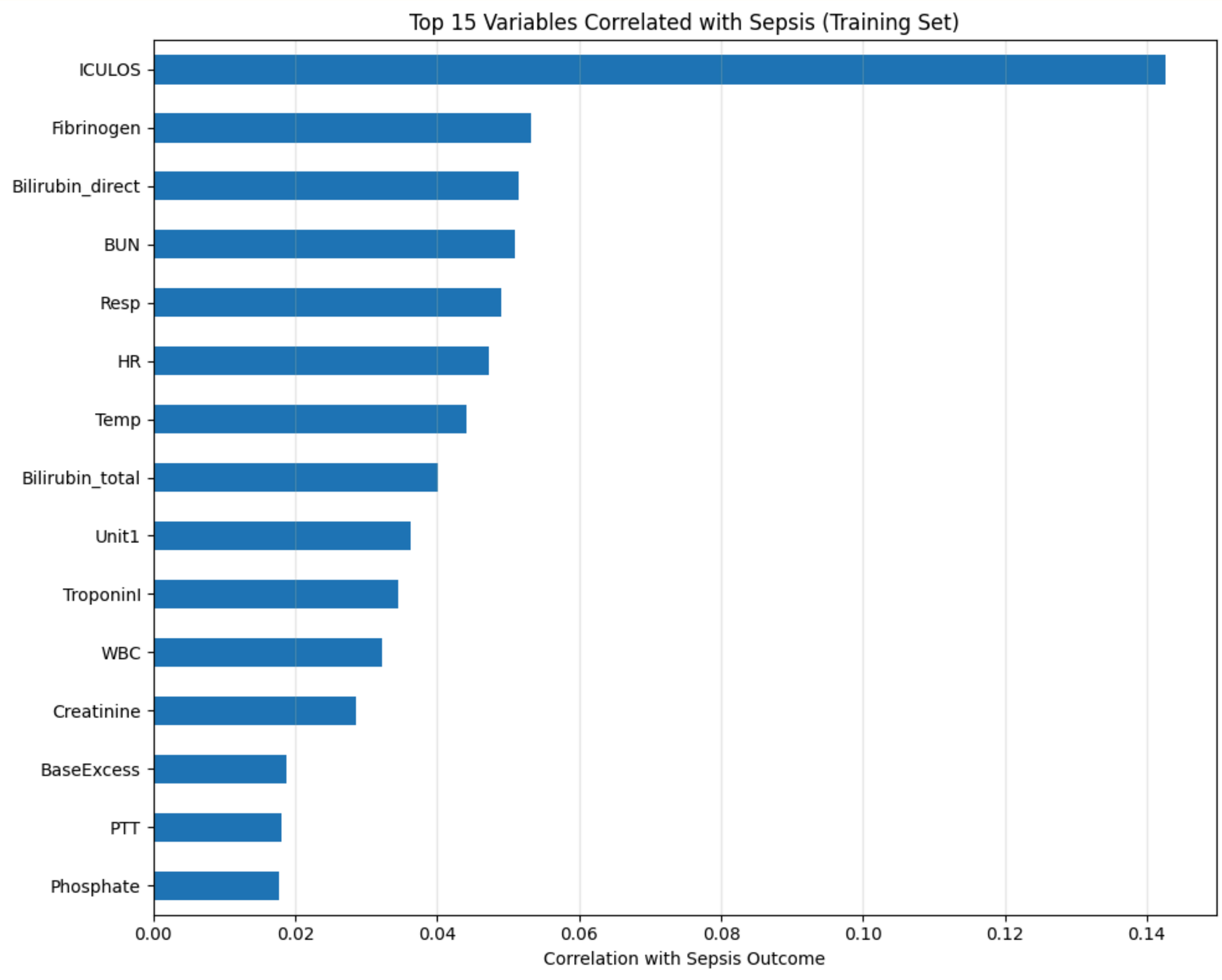


MISSING DATA



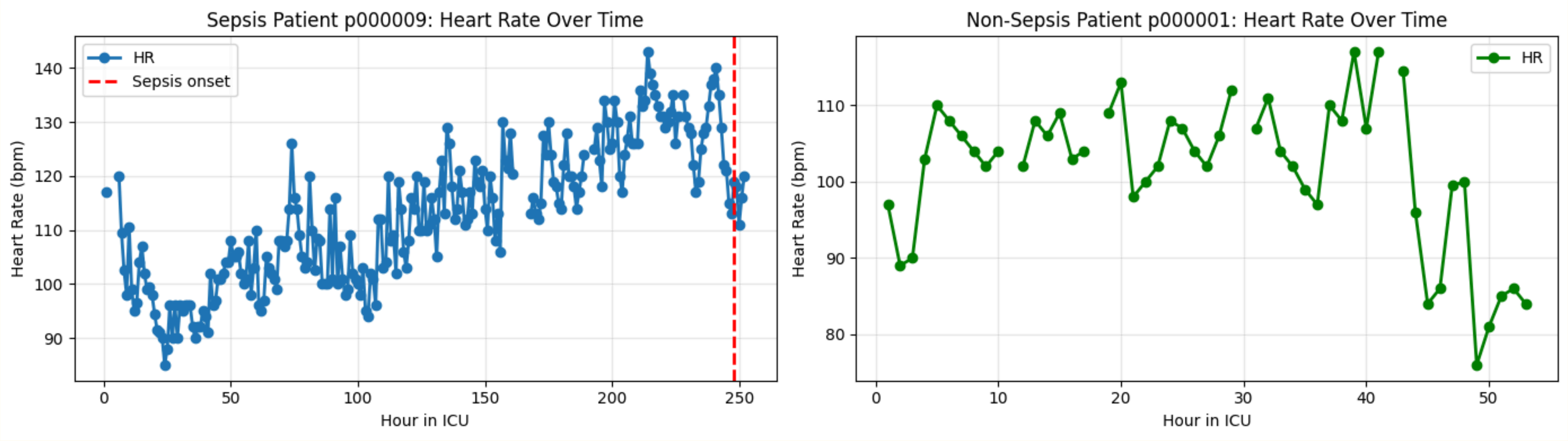
- Missing is NOT random; sepsis patients have more lab measurements
- Strategy → forward-fill + measurement indicator flags

SEPSIS CORRELATION



SEPSIS VS NON-SEPSIS TRAJECTORY

EX: HEART RATE OVER TIME



PREPROCESSING & FEATURE ENGINEERING

- Implemented forward-fill to address missing data
- Vital trends integral to sepsis detection
- Computed two new features for every clinical variable for each patient

1. six-hour rolling mean (ex: mean HR over last 6 hours)
2. three-hour difference (ex: HR changed by 10 in last 3 hours)

MODEL RESULTS AND COMPARISION

KEY METRICS

AUROC

- “Area under the Receiver Operating Characteristic Curve”
- How well model ranks sepsis cases higher than non-sepsis
- Best for balanced datasets
- 0 -1 scale
- 0.5 = random guessing
- 1 = perfect

PHYSIONET UTILITY SCORE

- How early the model catches sepsis
- Rewards predicting sepsis ≥3 hours early (+0.5 to +1.0 per patient)
- Penalizes missing sepsis entirely (-2.0 per patient)
- Result = average score across all patients

AUPRC

- “Area under the Precision-Recall Curve”
- How precise the positive predictions of a model are when the class is imbalanced
- Best for imbalanced datasets → more useful for this dataset
- 0 -1 scale
- Baseline = prevalence of the positive class (2% sepsis)
- 1 = perfect

SENSITIVITY & SPECIFICITY

- Sensitivity: "Of sepsis cases, how many do we catch?" (high is good for early warning)
- Specificity: "Of non-sepsis cases, how many do we correctly identify?" (high is good to avoid false alarms)

ADDRESSING CLASS IMBALANCE

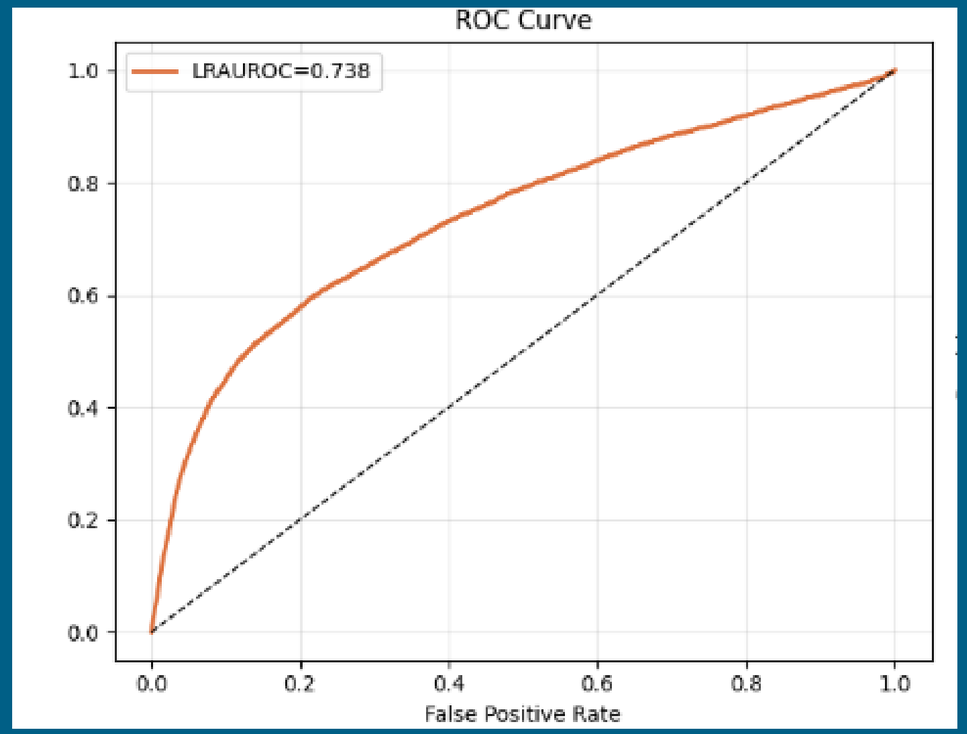
Strategy	How It Works	Impact
class_weight='balanced'	Weight sepsis cases 49× heavier in loss function	Forces model to care about minority class
AUPRC metric	Ignore accuracy; use precision-recall instead	Appropriate for imbalance (AUROC misleading)
PhysioNet utility score	Heavily penalize missed cases (-2.0)	Forces sensitivity over specificity
LSTM temporal modeling	Learn temporal patterns, not just raw values	Captures subtle signals that predict sepsis

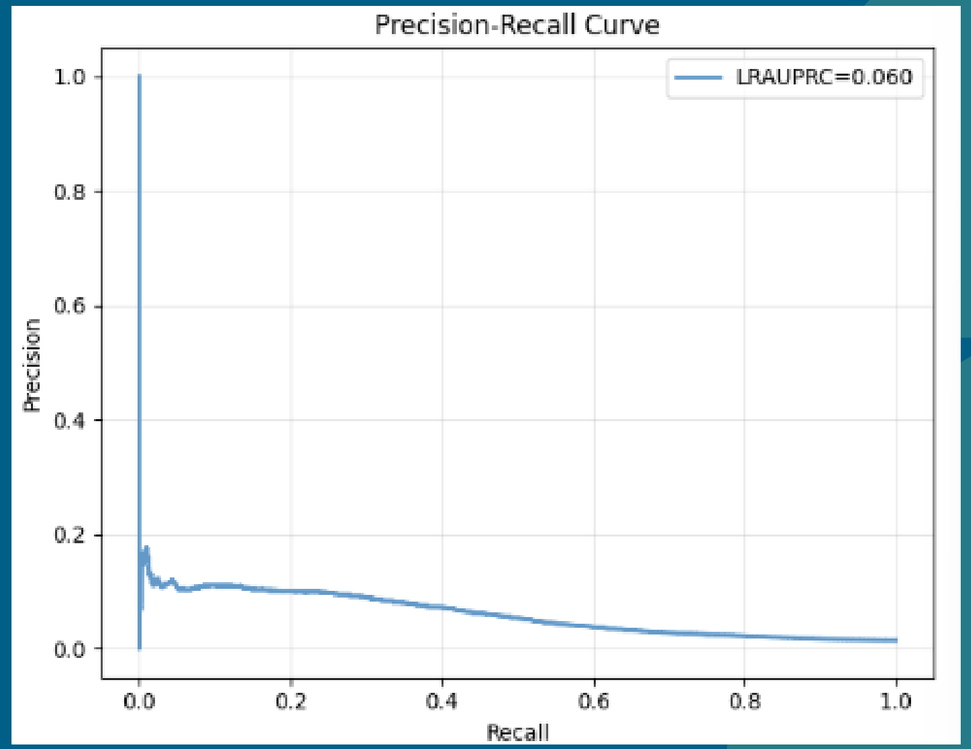
LOGISTIC REGRESSION

- Simple linear model
- Pros: fast to train, interpretable, established baseline
- Cons: treats each hour independently, doesn't catch temporal patterns

RESULTS

- AUROC: 0.7376 (decent)
- AUPRC: 0.0598 (good for imbalanced data)
- Utility score: -0.6741 (too many missed cases)
- Sensitivity: 84.1% (catches most sepsis)
- Specificity: 34.5% (many false alarms)





LSTM - TEMPORAL MODEL

- "Long Short-Term Memory" neural network that remembers patterns from past hours
- Learns sequences like: "HR rising + RR rising + temp falling = sepsis coming"
- Pros: captures temporal patterns for sepsis development, better at early detection
- Cons: blackbox, takes longer to train, more RAM intensive

Results:

- AUROC: 0.7498 (slightly better)
- AUPRC: 0.0556 (similar)
- Utility score: -0.1754 **(74% better than LR)**
- LSTM catches sepsis earlier than LR

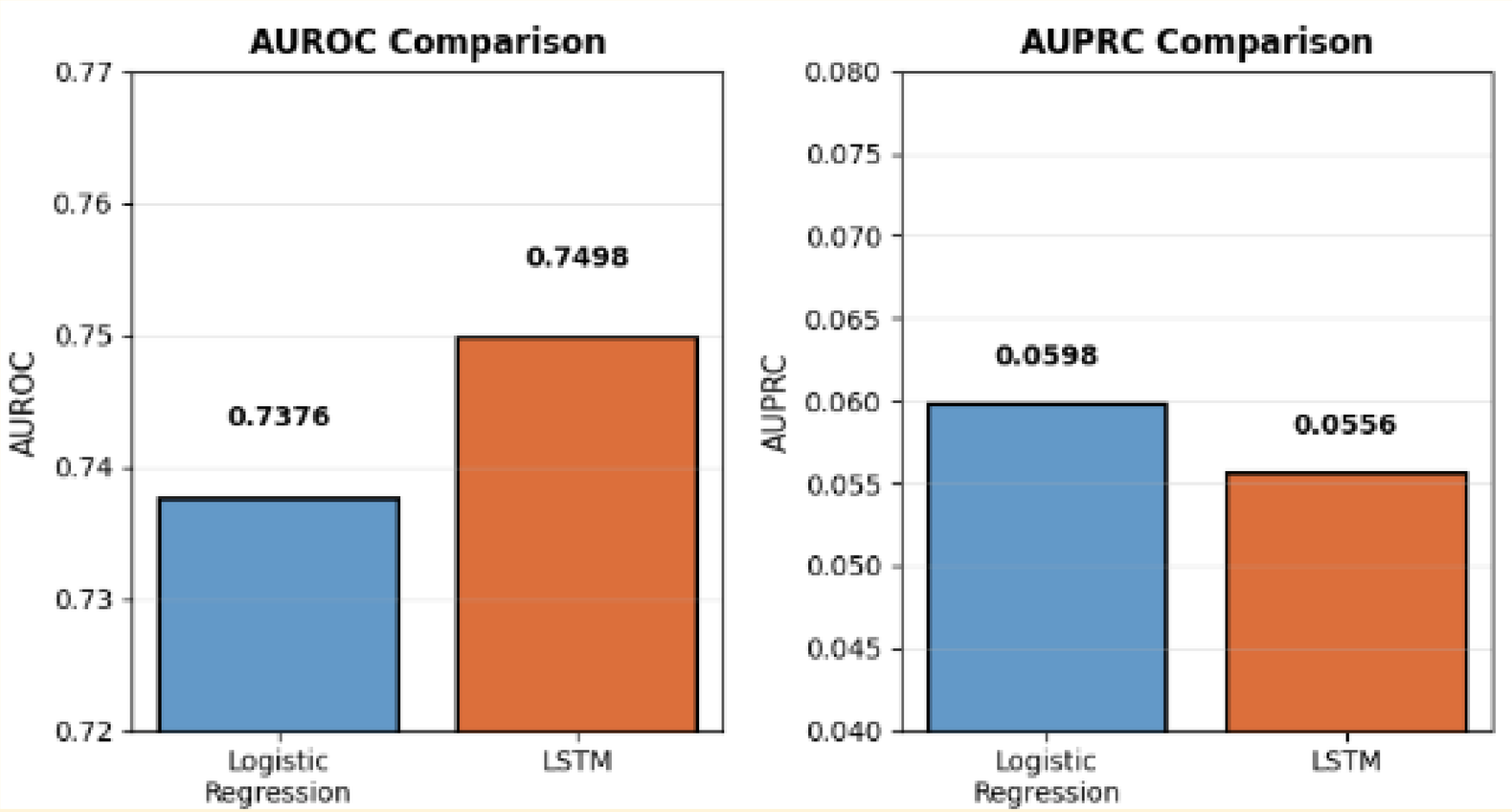
QSOFA

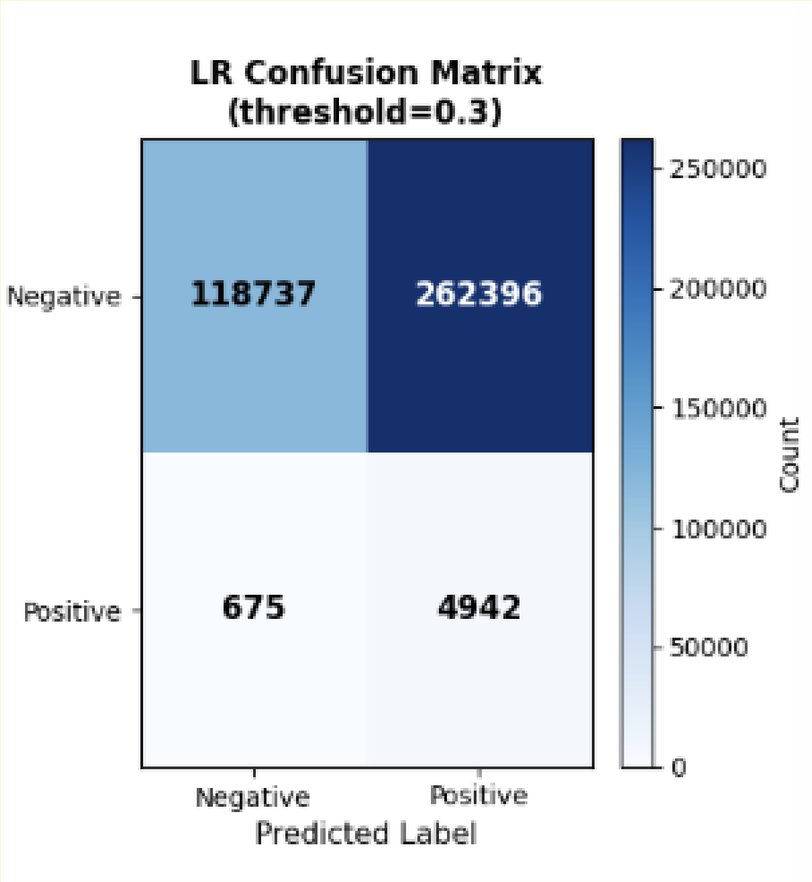
- "Quick Sequential Organ Failure Assessment"
- Bedside clinical scoring tool used to rapidly identify sepsis
  - Respiratory rate > 22 breaths/min? → 1 point
  - Score ≥ 2 = sepsis alert
- Pros: Interpretable, high specificity
- Cons: low sensitivity

Results:

- AUROC: 0.5701 (poor--barely better than random)
- AUPRC: 0.0188 (very poor)
- Sensitivity: 4.79% (misses 95% of sepsis!)
- Specificity: 98.6% (almost never alerts)

MODEL COMPARISIONS





MODEL COMPARISIONS

Metric	Your LR	Your LSTM	qSOFA
AUROC	0.7376	0.7498	0.5701
AUPRC	0.0598	0.0556	0.0188
Sensitivity	84.1%	~85%	4.79%
Specificity	34.5%	~33%	98.6%
Utility Score	-0.6741	-0.1754	(very negative)

CLINICAL DEPLOYMENT:

- LR → aggressive early warning system where missing a case is worse than a false alarm
- LSTM → Real-time ICU monitoring where early detection is critical
- qSOFA → Rule-in confirmation only (if qSOFA positive, patient definitely has sepsis)

CLINICAL IMPLICATIONS

- Recommended model → LSTM
  - Best utility score and high sensitivity at the cost of higher false alarms
- qSOFA could complement LSTM by providing a rule-in confirmation
- Cost-effective model → missed sepsis cases are higher cost
- Mitigation of alert fatigue requires smart EHR integration
  - Only alert if sepsis risk >0.4 and the patient is high risk
  - Accept only top 50% of predictions

LIMITATIONS

- Physical limitations: density of dataset and limited RAM
- Sepsis labels based on clinical documentation
  - Detection lag
  - Ascertainment bias
  - Prospective gap
- Negative utility score
  - Reflects the difficulty of the problem, not model failure
  - Positive utility score nearly impossible with current data (rare disease, weak signals, unavoidable false alarms)

FUTURE WORK

- Proper integration into clinical workflow → adjust thresholds to limit alert fatigue from false alarms
- Develop a standardized method for consistent data collection
  - Prospective data collection
  - Validation against microbiologic confirmation
  - Adjustment for documentation delays